

WELL COMPLETION REPORT (SYNTHETIC)

OIL-RDS-012 - Rudrasagar-012

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000138
Field / District	Rudrasagar / Sivasagar
Block	PEL-ASSAM-06
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2022-04-07 / 2022-07-07
Measured / TVD	2752 m / 2715 m
Trajectory	Vertical; inc 3.4 deg; az 99.7 deg
Formation Interval	1624-2551 m
Lithology	Fine sandstone with carbonaceous shale
Pore Pressure Reference	3402 psi
Fracture Gradient	15.01 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (52.3/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	688	Surface
9 5/8 in casing	1596	Intermediate
7 in liner/casing	2421	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2022-04-25 676 m	Pressure Spike	Critical	Circulating pressure climbed without equivalent torque change. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2022-05-05 992 m	Cementing Issue	Critical	Cement returns were below planned percentage. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.
2022-05-11 1155 m	Mud Loss	Critical	Active pit volume declined during stable pump rate. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2022-04-23 638 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2022-04-27 733 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2022-05-11 1173 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.
2022-05-31 1736 m	Torque Spike	High	Torque rose without corresponding pressure anomaly. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.

Completion Summary

The synthetic well reached 2752 m MD in the Barail interval. The NWIS knowledge record links 17 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.